

For metals, the thermal conductivity is mainly a function of the motion of free electrons. As the temperature increases, the molecular vibrations increase (in turn increasing the mean free path of molecules). So, they obstruct the flow of free electrons, thus reducing the conductivity.

In case of non-metals, there are no free electrons. So, only the molecular vibrations are responsible for conduction of heat and hence for non-metals, the conductivity increases with increase in temperature.

3. The characteristic of superconductor is –

- (a) High permeability (b) Low permeability
(c) Zero permeability (d) Infinite permeability

39th B.P.S.C. (Pre) 1994

Ans. (c)

In electromagnetism, permeability is a property of a material that describes the ability of a material to support the formation of a magnetic field within itself or we can say that it is the degree of magnetization that a material obtains in response to an applied magnetic field. It is represented by Greek letter μ (Mu). The permeability and electrical resistance of the superconductor is zero.

4. On heating, the resistance of a semiconductor :

- (a) increases
(b) decreases
(c) remains same
(d) first increases and then decreases
(e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (b)

With the increase in temperature, the conductivity of the semiconductor material increases. As with increase in temperature, outermost electrons acquire energy and hence by acquiring energy, the outermost electrons leave the shell of the atom. Hence, with an increase in temperature, number of carriers in the semiconductor material increases which leads to increase in the conductivity of the material. So we can say that the semiconductor material has negative temperature coefficient i.e. with an increase in temperature, resistance decreases.

5. Which one of the following metals is used as semiconductor in transistors?

- (a) Copper (b) Germanium
(c) Graphite (d) Silver

44th B.P.S.C. (Pre) 2000

Ans. (b)

Germanium and Silicon are the main elements which are used as a semiconductor in transistors. Their conductivity on the normal/room temperature lies between the conductivity of conductors and insulators.

Nuclear Physics

1. The modern atomic symbols are based on the method proposed by

- (a) Bohr (b) Berzelius
(c) Dalton (d) Thomson
(e) None of the above / More than one of the above

B.P.S.C. CDPO 2022

Ans. (b)

Berzelius suggested that the initial letter of an element written in capitals should represent that particular element. This method suggested by him laid the basis of the IUPAC system of chemical symbols and formulae.

2. The nucleus of an atom consists of –

- (a) electrons and neutrons
(b) electrons and protons
(c) protons and neutrons
(d) electrons only
(e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre) 2021

Ans. (c)

Atomic nucleus is made up of protons and neutrons. The mass of the nucleus is made up of protons and neutrons and the positive nucleus charge is due to protons only. Protons and neutrons are together called nucleons because the particles are components of the atomic nucleus.

3. The components of atomic nucleus are :

- (a) Electron and proton
(b) Electron and neutron
(c) Protons and neutrons
(d) Protons, neutrons and electrons

44th B.P.S.C. (Pre) 2000-01

Ans. (c)

See the explanation of the above question.

4. Which among the following is a positively charged particle emitted by a radioactive element?

- (a) Cathode ray (b) Beta ray
(c) Alpha ray (d) More than one of the above
(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (c)

67th B.P.S.C. (Pre) 2021

Positively charged particles emitted by a radioactive elements are called alpha ray/particles. Alpha particles are composite particles consisting of two protons and two neutrons bound together.

The emission of alpha particles from radioactive elements is a significant form of radiation. However, Alpha particles have relatively low penetration power.

Common Alpha-emitting radioactive elements include uranium-238, radium-226 & plutonium-239.

5. Which of the following is not a radioactive element?

- (a) Uranium
- (b) Radium
- (c) Thorium
- (d) Cadmium
- (e) None of the above/More than one of the above

B.P.S.C. CDPO 2018

Ans. (d)

Cadmium is not a radioactive element. Cadmium is found in nature in mineral form. Cadmium rods are used in the nuclear reactor to slow down the speed of neutron.

6. Which of the following is a light element indicating radioactivity?

- (a) Hydrogen
- (b) Deuteron
- (c) Tritium
- (d) Helium
- (e) None of the above / More than one of the above

B.P.S.C. Auditor 2021

Ans. (c)

Tritium is the only radioactive isotope of hydrogen and the other two have stable nuclei. As hydrogen is the lightest element in the periodic table, tritium will be the lightest radioactive isotope in the periodic table.

7. The energy of Sun is released due to-

- (a) Nuclear Fission
- (b) Nuclear Fusion
- (c) Oxidation Reactions
- (d) Reduction Reactions

42nd B.P.S.C. (Pre) 1997

Ans. (b)

The Sun produces energy by the nuclear fusion in its core. Since there is a huge amount of hydrogen in the core, these atoms stick together and fuse into a helium atom. This energy is then radiated out from the core and moves across the solar system.

8. Hydrogen bomb based on the principle of :

- (a) Controlled fusion reaction
- (b) Uncontrolled fusion reaction
- (c) Controlled fission reaction
- (d) Uncontrolled fission reaction
- (e) None of the above / more than one of the above

Ans. (b)

The hydrogen bomb is based on an uncontrolled nuclear fusion reaction, in which two smaller nuclei fuse to form a larger nucleus, releasing enormous amounts of energy.

9. The principle of atomic bomb is based on –

- (a) Nuclear fusion
- (b) Nuclear fission
- (c) Above both
- (d) Above none

39th B.P.S.C. (Pre) 1994

Ans. (b)

The principle of atomic bomb is based on nuclear fission. In this reaction the nucleus of an atom splits into two or more smaller nuclei. As a result of this fission, energy is released.

10. Which metal can be used for producing electricity?

- (a) Uranium
- (b) Iron
- (c) Copper
- (d) Aluminium

47th B.P.S.C. (Pre) 2005

Ans. (a)

The main use of uranium in the civilian sector is to fuel nuclear power plants. 1 kg of U-235 can theoretically produce about 20 Tera joules of energy which is equivalent to 1500 tons of coal.

11. Radioactive substance emits –

- (a) Alpha rays
- (b) Beta rays
- (c) Gamma rays
- (d) All the above

40th B.P.S.C. (Pre) 1995

Ans. (d)

Radioactive elements are elements that have an unstable nucleus. When the nuclei are considered unstable, they radiate alpha, beta and gamma radiation and are converted into a stable element. This type of radiation is invisible to the naked eye.

12. Fast breeder reactor is a nuclear reactor which produces electricity –

- (a) By fission process
- (b) With the help of solar cells
- (c) By burning phenyl substances used by it and by its reproduction
- (d) By using refined hard water

38th B.P.S.C. (Pre) 1992

Ans. (a)

A fast breeder reactor is a nuclear reactor which produces electricity by the fission process. This type of reactor produces more fissile material than it burns.

13. Which of the following is the correct match?

- (a) Electric motor → Electrical energy to mechanical